

Advanced Machine Learning

DATA 642

Lecture 7 — Multi-modal learning

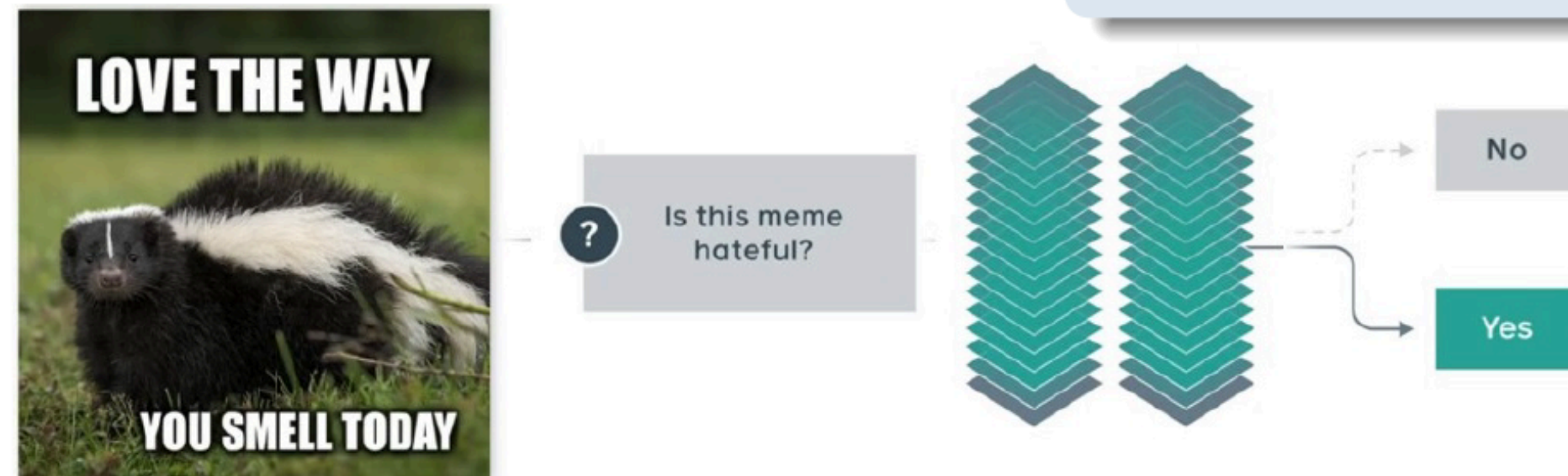
*Figures and notes are from the paper Multimodal Machine Learning: A Survey and Taxonomy by Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency. DOI: [10.1109/TPAMI.2018.2798607](https://doi.org/10.1109/TPAMI.2018.2798607).

Significance of multi-modal learning



To explore the underlying structure of **multi-modal datasets**, i.e., information extraction by integrating and modeling **multiple modalities**.

Understand **content holistically**.



Challenges in Multimodal Fusion

Representation:

- Learning to represent and summarize multimodal data effectively to leverage complementarity and redundancy.
- Multimodal data's heterogeneity complicates constructing such representations.
- Example: Language as symbolic, audio and visual as signals.

Translation:

- Mapping data from one modality to another.
- Data heterogeneity and open-ended or subjective inter-modality relationships pose difficulties.
- Example: Multiple correct descriptions for an image; no single perfect translation.

Alignment:

- Identifying direct relations between elements from different modalities.
- Need to measure similarity, handle long-range dependencies, and resolve ambiguities.
- Example: Aligning recipe steps with a cooking video.

Challenges in Multimodal Fusion (cont.)

Fusion:

- Integrating information from multiple modalities for prediction.
- Example: Audio-visual speech recognition, where lip motion description is fused with speech signal.
- Challenges include varying predictive power, noise topology, and potentially missing data.

Co-learning:

- Transferring knowledge between modalities, representations, and predictive models.
- Techniques include co-training, conceptual grounding, and zero-shot learning.
- Relevant when one modality has limited resources (e.g., annotated data).

Multimodal Representations

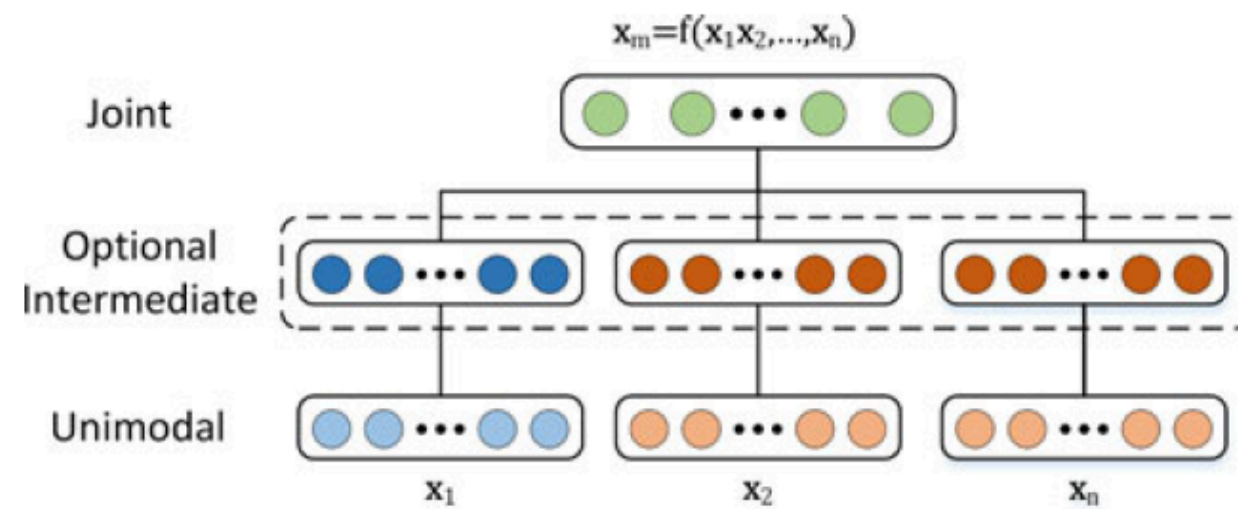
Representing data in a format that a computational model can work with has always been a challenge in machine learning. This section discusses the complexities and advancements in multimodal representations.

Fundamentals of Representations

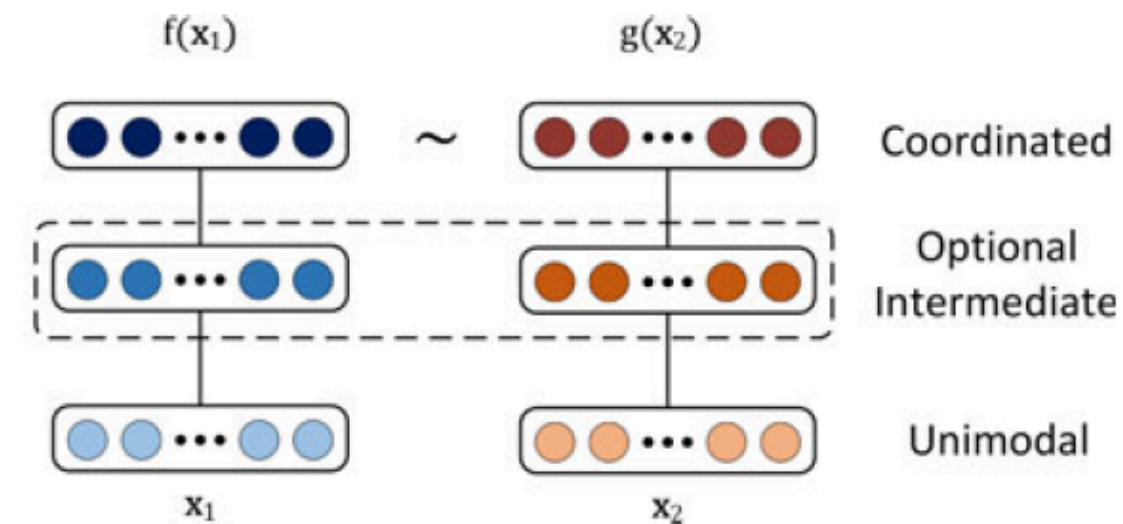
- Multimodal representation refers to a representation of data using information from multiple entities, each represented by a vector or tensor.
- Challenges include combining heterogeneous data, dealing with varying noise levels, and handling missing data.
- Good representations are crucial for machine learning model performance, impacting areas like speech recognition and object classification.
- Desired properties of representations include smoothness, coherence, sparsity, and natural clustering.
- Additional properties for multimodal representations include reflecting similarity between concepts, ease of obtaining representations, and ability to handle missing modalities.

Multimodal Representation Types

- Two categories proposed: joint and coordinated representations.
- Joint representations combine modalities into the same space, while coordinated representations process modalities separately but enforce similarity constraints.
- Joint representations expressed as mathematical functions combining unimodal representations.
- Coordinated representations enforce similarity or structure between modalities' representations.



(a) Joint representation

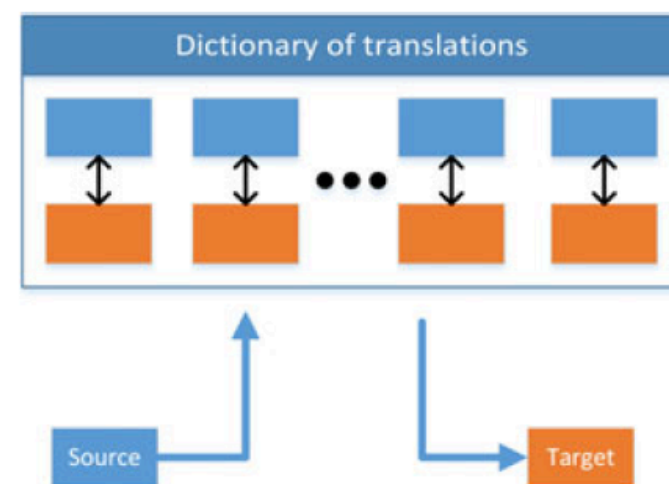


(b) Coordinated representations

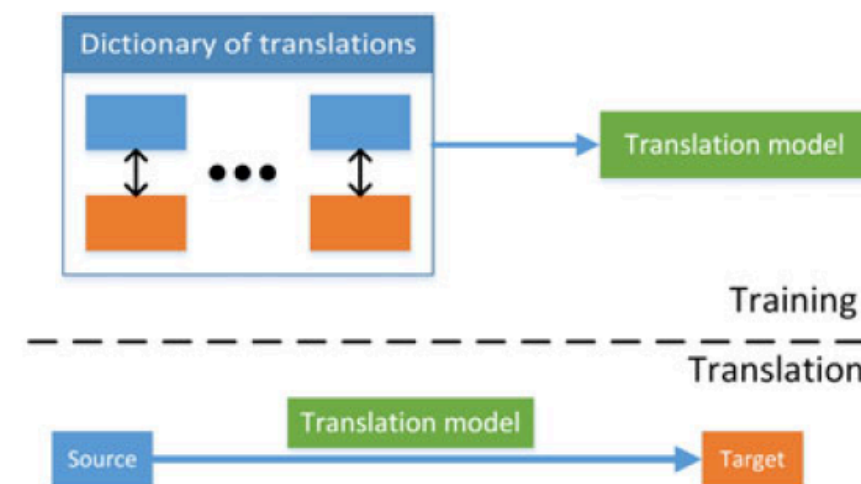
Translation

Translation focusing on translating between different modalities, such as generating text from images or vice versa.

- **Example-based models** rely on dictionaries for translation, where retrieval-based models find the closest match in the dictionary, while combination-based models combine examples for better translation. Generative models construct models to generate translations, which is more challenging as it requires generating sequences of symbols. Grammar-based, encoder-decoder, and continuous generation models are discussed within generative approaches.
- **Grammar-based models** use predefined templates and restricted grammars, while encoder-decoder models encode the source modality and decode to generate the target modality. Continuous generation models produce outputs at each time step, suitable for sequence-to-sequence translation tasks.



(a) Example-based



(b) Generative

Multimodal Alignment Summary

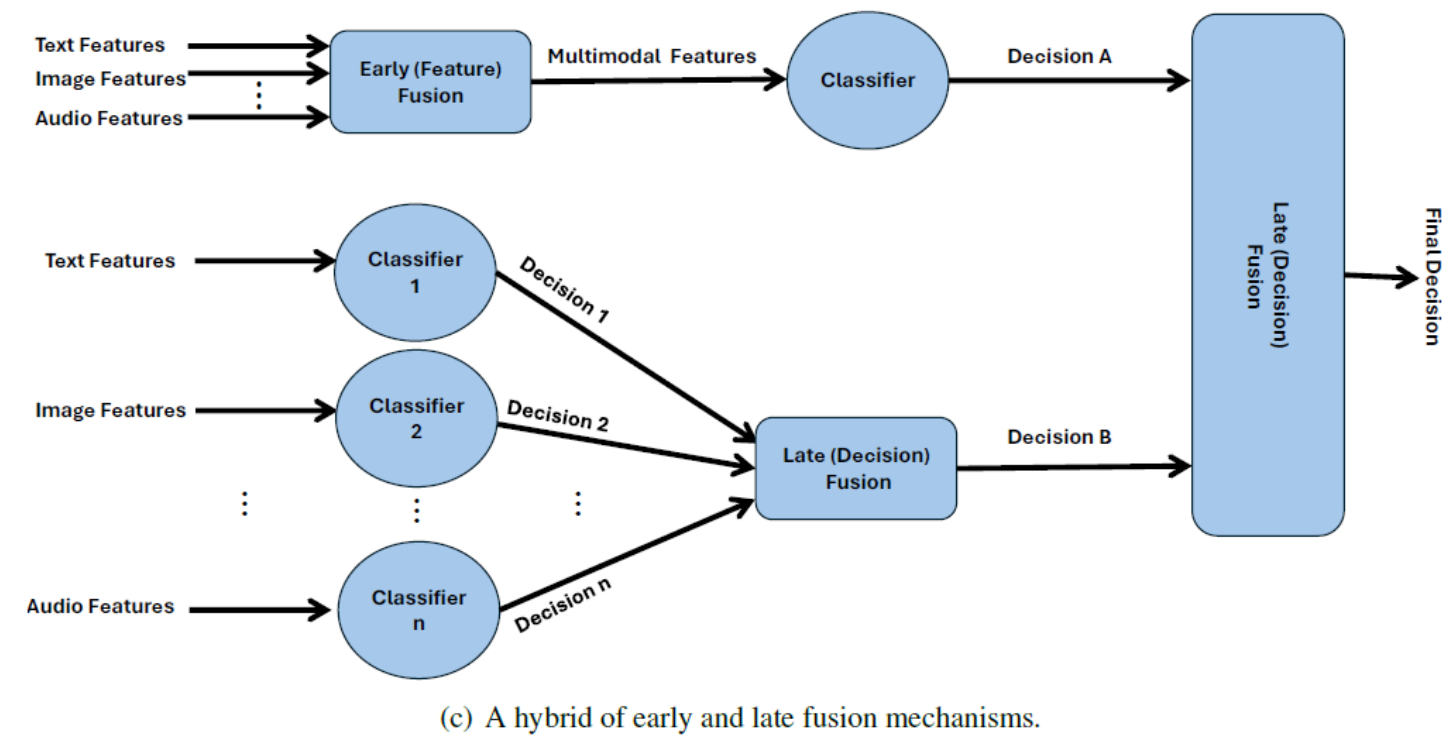
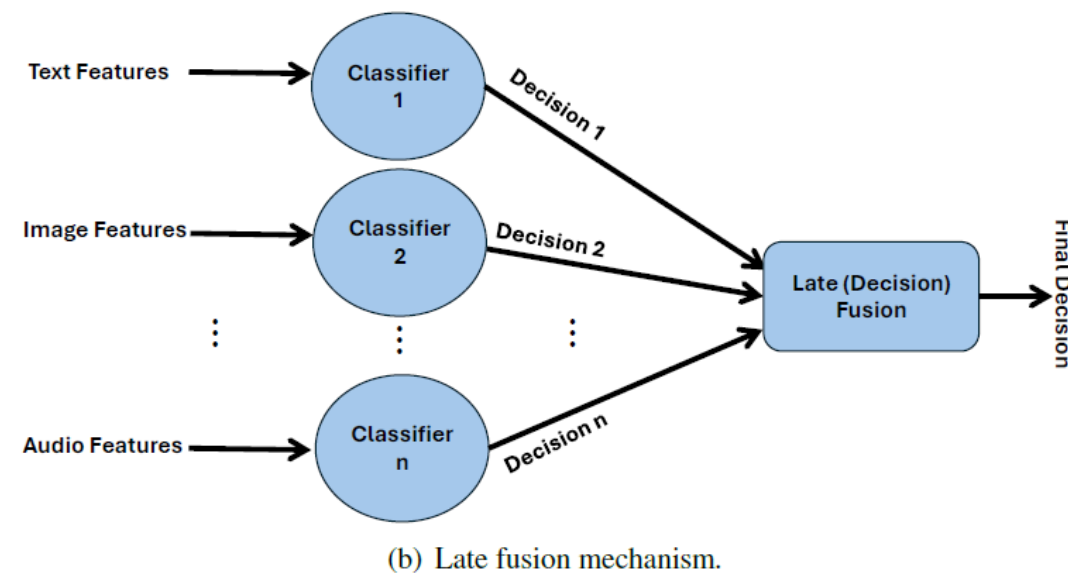
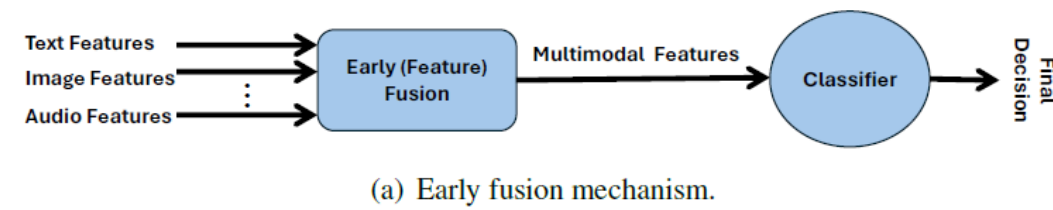
Finding relationships between sub-components of instances from different modalities.

- **Types:**
 - *Explicit:* Aligning sub-components explicitly.
 - *Implicit:* Used as an intermediate step for other tasks.
- **Explicit Alignment:**
 - Unsupervised approaches like DTW and graphical models.
 - Supervised methods relying on labeled instances.
 - Deep learning methods using CNNs and LSTMs.
- **Implicit Alignment:**
 - Early work with graphical models.
 - Modern neural network approaches, including attention mechanisms.
- **Applications:** Multimedia retrieval, translation, question-answering.

Multi-modal fusion

Types of Model-Agnostic Fusion:

- *Early Fusion*: Integrates features immediately after extraction.
- *Late Fusion*: Integration after each modality makes a decision.
- *Hybrid Fusion*: Combines outputs from early fusion and unimodal predictors.



Co-learning

Enhancing modeling of a resource-poor modality using knowledge from a resource-rich modality. Particularly useful when one modality lacks resources (e.g., annotated data, reliable labels).

Approaches:

- *Parallel*: Require directly linked observations from different modalities.
- *Non-parallel*: Do not need direct links, often utilize overlap in categories.
- *Hybrid*: Bridging modalities through a shared modality or dataset.

Usage:

- Typically helper modality is used only during training, not during testing.

